

DATA CLEANING REPORT

E-Commerce Sales Dataset — Day 1

7-Day Data Analyst Portfolio Roadmap — Full Step-by-Step Documentation

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1. Introduction

This report documents, in detail, every step of the data cleaning process applied to the e-commerce transaction dataset (raw_sales_data.csv) using Microsoft Excel, as part of Day 1 in a Data Analyst portfolio roadmap. The raw dataset contains 4,100 transaction rows spanning January 2023 to December 2024, with a number of data quality issues deliberately designed to mirror real-world data conditions: duplication, missing values, inconsistent formatting, incorrect data types, and outliers.

The goal of this stage is to produce a clean, consistent dataset ready for the subsequent stages (Excel analysis, SQL, Python, and the Tableau dashboard), so that the figures reported across the whole roadmap remain consistent with one another. Every step below is accompanied by the formula used and a screenshot as process evidence, rather than a narrative summary alone, so it can serve as a reference to revisit without having to recall everything from scratch.

2. Executive Summary

Table 1 summarizes how the dataset’s condition changed before and after the cleaning process.

Metric	Before	After
Row count	4,100	3,933
Column count	11	12 (+ revenue)
Duplicate rows	99	0
Missing values (all columns)	533 cells	0
Extreme outliers (quantity & price)	27 rows	0
order_date format	5 mixed formats	1 format (YYYY-MM-DD)

Table 1. Comparison of dataset condition before and after cleaning

3. Data Cleaning Steps

3.1 Removing Duplicate Data

Duplicate rows were identified using the Data > Remove Duplicates feature with all columns checked, so that only rows identical across every column (including order_id) were treated as duplicates — not merely partial similarity. This approach avoids the mistake of deleting two distinct transactions that happen to share the same product or date.

Step 1 — Identify duplicate rows with a row signature

To make sure only truly identical rows get flagged, every column is concatenated into a single "row signature" using TEXTJOIN, then identical signatures are highlighted with Conditional Formatting > Highlight Duplicate Values:

```
=TEXTJOIN("|",TRUE,A2:K2)
```

	D	E	F	G	H	I	J	K	L
	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	row_signature
1	Andi Halim		Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card	ORD100016 45253 CUST1353 Andi Halim Sports & Outdoors Water Bottle 2 28.97 Completed Credit Card
2	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	ORD102704 44999 CUST1478 Lisa Kurniawan East Toys Building Blocks 2 46.3 Returned Credit Card
3	Ahmad Halim	Mid-West	Beauty	Moisturizer	1	35.09	Completed	Gift Card	ORD103490 45392 CUST1532 Ahmad Halim Mid-West Beauty Moisturizer 1 35.09 Completed Gift Card
4	Ayu Hidayat	SOUTH	Sports & Outdoors	Camping Tent	2	12.94	Completed	Debit Card	ORD101567 45526 CUST1390 Ayu Hidayat SOUTH Sports & Outdoors Camping Tent 2 12.94 Completed Debit Card
5	James Yusuf	SOUTH	Sports & Outdoors	Hiking Backpack	1	23.12	Completed	PayPal	ORD101896 45343 CUST1563 James Yusuf SOUTH Sports & Outdoors Hiking Backpack 1 23.12 Completed PayPal
6	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	ORD101473 45522 CUST1052 Fajar Halim Midwest Beauty Sunscreen SPF50 1 33.88 Completed Gift Card
7	John Tan	South	Books	Fiction Novel	2	10.88	Completed	Gift Card	ORD101311 45519 CUST1728 John Tan South Books Fiction Novel 2 10.88 Completed Gift Card
8	Kevin Anderson	WEST	Beauty	Moisturizer	1	21.46	Completed	PayPal	ORD100537 45119 CUST1445 Kevin Anderson WEST Beauty Moisturizer 1 21.46 Completed PayPal
9	John Yusuf	MIDWEST	Home & Kitchen	Cutting Board	1	102.68	Completed	PayPal	ORD101433 45040 CUST1699 John Yusuf MIDWEST Home & Kitchen Cutting Board 1 102.68 Completed PayPal
10	Ayu Smith	South	Sports & Outdoors	Resistance Bands	1	56.4	Completed	PayPal	ORD101351 45149 CUST1708 Ayu Smith South Sports & Outdoors Resistance Bands 1 56.4 Completed PayPal
11	Nina Lee	Midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	ORD102767 45191 CUST1868 Nina Lee Midwest Books Business Strategy 1 18.14 Completed Credit Card
12	John Smith	South	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	ORD101388 45366 CUST1211 John Smith South Clothing Winter Jacket 2 35.68 Completed Gift Card
13	Fajar Santos	East	Home & Kitchen	Storage Containers	1	112.68	Completed	Gift Card	ORD101780 45500 CUST1336 Fajar Santos East Home & Kitchen Storage Containers 1 112.68 Completed Gift Card
14	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	ORD101857 45143 CUST1104 Sarah Tan East Electronics Webcam 1 68.92 Completed Debit Card
15	Lisa Taylor	MIDWEST	Beauty	Shampoo	2	23.18	Cancelled		ORD100800 45114 CUST1352 Lisa Taylor MIDWEST Beauty Shampoo 2 23.18 Cancelled
16	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	ORD101695 45432 CUST1225 Budi Kurniawan West Clothing Hoodie 2 66.21 Returned PayPal
17	Linda Chen	East	Sports & Outdoors	Camping Tent	1	Cancelled		PayPal	ORD102093 45316 CUST1824 Linda Chen East Sports & Outdoors Camping Tent 1 Cancelled PayPal
18	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	ORD101296 45571 CUST1003 Siti Anderson East Electronics Power Bank 3 134.85 Completed PayPal
19	Siti Wilson	Midwest	Sports & Outdoors	Yoga Mat	1	75.34	Completed	Credit Card	ORD101842 45386 CUST1741 Siti Wilson Midwest Sports & Outdoors Yoga Mat 1 75.34 Completed Credit Card
20	Nina Garcia	West	Sports & Outdoors	Hiking Backpack	1	49.79	Returned	Credit Card	ORD101291 45225 CUST1804 Nina Garcia West Sports & Outdoors Hiking Backpack 1 49.79 Returned Credit Card
21	Robert Smith	West	Clothing	Denim Jeans	4	62.83	Completed	Debit Card	ORD102554 45157 CUST1210 Robert Smith West Clothing Denim Jeans 4 62.83 Completed Debit Card
22	Siti Tan	EAST	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	ORD101824 45495 CUST1324 Siti Tan EAST Beauty Face Mask Set 1 39.05 Completed Debit Card
23	David Wijaya	EAST	Home & Kitchen	Blender	4	115.34	Completed	Debit Card	ORD101777 45654 CUST1227 David Wijaya EAST Home & Kitchen Blender 4 115.34 Completed Debit Card
24	David Anderson	South	Sports & Outdoors	Camping Tent	1	30.3	Completed	Debit Card	ORD102224 45051 CUST1588 David Anderson South Sports & Outdoors Camping Tent 1 30.3 Completed Debit Card
25	Sarah Johnson	South	Clothing	Hoodie	1	46.19	Completed	Debit Card	ORD101815 45477 CUST1265 Sarah Johnson South Clothing Hoodie 1 46.19 Completed Debit Card
26	Dewi Sari	EAST	Books	Self-Help Guide	1	29.66	Completed	Credit Card	ORD101819 44966 CUST1343 Dewi Sari EAST Books Self-Help Guide 1 29.66 Completed Credit Card
27	Putri Halim	East	Sports & Outdoors	Camping Tent	1	72.43	Cancelled	PayPal	ORD101076 45124 CUST1680 Putri Halim East Sports & Outdoors Camping Tent 1 72.43 Cancelled PayPal
28	Maria Wilson	SOUTH	Toys	Board Game	1	58.15	Completed	PayPal	ORD101910 45024 CUST1887 Maria Wilson SOUTH Toys Board Game 1 58.15 Completed PayPal
29	Dewi Tan	East	Beauty	Lip Balm	2	7.03	Completed	PayPal	ORD100326 45480 CUST1182 Dewi Tan East Beauty Lip Balm 2 7.03 Completed PayPal
30	Nina Garcia	MIDWEST	Toys	RC Car	1	5.57	Completed		ORD101328 45032 CUST1254 Nina Garcia MIDWEST Toys RC Car 1 5.57 Completed
31	Sarah Lee	East	Clothing	Cotton T-Shirt	1	54.55	Completed	Debit Card	ORD102634 45070 CUST1757 Sarah Lee East Clothing Cotton T-Shirt 1 54.55 Completed Debit Card
32	Rina Taylor	SOUTH	Beauty	Moisturizer	1	25.47	Completed	Debit Card	ORD103616 45348 CUST1128 Rina Taylor SOUTH Beauty Moisturizer 1 25.47 Completed Debit Card
33	Ahmad Hidayat	Mid-West	Home & Kitchen	Coffee Maker	2	52.50	Cancelled	Gift Card	ORD101919 45245 CUST1308 Ahmad Hidayat Mid-West Home & Kitchen Coffee Maker 2 52.50 Cancelled Gift Card
34	Siti Wilson	WEST	Beauty	Shampoo	2	24.48	Completed	PayPal	ORD101226 45042 CUST1206 Siti Wilson WEST Beauty Shampoo 2 24.48 Completed PayPal
35	Sarah Chen	WEST	Toys	Board Game	1	58.26	Completed	Debit Card	ORD102507 45192 CUST1645 Sarah Chen WEST Toys Board Game 1 58.26 Completed Debit Card
36	Rina Anderson	WEST	Toys	Puzzle 1000pc	1	25.46	Completed	Debit Card	ORD100687 44943 CUST1370 Rina Anderson WEST Toys Puzzle 1000pc 1 25.46 Completed Debit Card
37	Rudi Chen	South	ELECTRONICS	Bluetooth Speaker	2	93.44	Completed	PayPal	ORD101423 45047 CUST1838 Rudi Chen South ELECTRONICS Bluetooth Speaker 2 93.44 Completed PayPal
38	Kevin Lee	East	Clothing	Running Shorts	2	24.84	Completed	PayPal	ORD101033 45560 CUST1753 Kevin Lee East Clothing Running Shorts 2 24.84 Completed PayPal
39	Putri Anderson	West	Clothing	Hoodie	2	Completed		PayPal	ORD102841 17/09/2023 CUST1635 Putri Anderson West Clothing Hoodie 2 Completed PayPal
40	Nina Garcia	SOUTH	Beauty	Moisturizer	2	10.48	Completed	Gift Card	ORD100687 44993 CUST1887 Nina Garcia SOUTH Beauty Moisturizer 2 10.48 Completed Gift Card
41	Ayu Chen	South	ELECTRONICS	Power Bank	2	142.03	Completed	Debit Card	ORD100619 45464 CUST1454 Ayu Chen South ELECTRONICS Power Bank 2 142.03 Completed Debit Card
42	Dewi Lee	Midwest	Beauty	Sunscreen SPF50	2	15.43	Completed	Debit Card	ORD101200 45232 CUST1608 Dewi Lee Midwest Beauty Sunscreen SPF50 2 15.43 Completed Debit Card
43	Maria Anderson	South	Toys	Puzzle 1000pc	1	19.88	Completed	Debit Card	ORD102075 45581 CUST1794 Maria Anderson South Toys Puzzle 1000pc 1 19.88 Completed Debit Card
44	Ayu Pratama	East	Sports & Outdoors	Yoga Mat	1	38.17	Completed	PayPal	ORD101211 24/03/2024 CUST1073 Ayu Pratama East Sports & Outdoors Yoga Mat 1 38.17 Completed PayPal
45	Sarah Tan	MIDWEST	Clothing	Cotton T-Shirt	4	38.53	Returned		ORD102968 44983 CUST1381 Sarah Tan MIDWEST Clothing Cotton T-Shirt 4 38.53 Returned

Figure 1. The row_signature column — rows with an identical signature are highlighted, indicating genuine duplicates

Step 2 — Remove Duplicates

Once the duplicate rows were verified, Data > Remove Duplicates was run with all columns checked:

The screenshot shows the Excel interface with the 'raw_sales_data' worksheet. The 'Remove Duplicates' task has been completed, resulting in 4,002 unique rows. The status bar at the bottom confirms that 98 duplicate values were found and removed, leaving 4,002 unique values.

Figure 2. Remove Duplicates result — 98 duplicate rows found and removed, leaving 4,002 unique rows

- Initial result: 98 exact-duplicate rows removed, leaving 4,002 unique rows.

Step 3 — Re-verification (Day 2)

Before the dataset was used for the analysis stage, it was re-checked with the same technique (row signature + color filter). One additional duplicate row was found (order_id ORD101170, customer Fajar Wijaya — Wool Socks, qty 3) that had slipped through the initial process:

The screenshot shows two identical rows in the Excel data, identified as order_id ORD101170. Both rows represent an order for 'Wool Socks' by 'Fajar Wijaya'.

Figure 3. Re-verification on Day 2 found 1 additional duplicate that slipped through the initial cleanup (rows 441 and 3935, identical content)

- Final total: 99 exact-duplicate rows removed (98 in the initial cleanup + 1 found during re-verification).

3.2 Standardizing the Category & Region Columns

Both columns had inconsistent capitalization ("electronics", "ELECTRONICS"), extra whitespace, and misspellings ("Cloths", "Electronic", "Mid-West"). Standardization was done through a helper column using PROPER + TRIM, followed by manual typo correction via Find & Replace, then the result was pasted back (Paste Special > Values) into the original column:

```
=PROPER(TRIM([@category]))
```

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	category_clean	
1	ORD100016	11/23/2023	CUST1353	Andi Halim		Sports & Ou	Water Bottle	2	28.97	Completed	Credit Card	Sports & Outdoors	
2	ORD102704	3/14/2023	CUST1478	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	F3))	
4	ORD103490	4/10/2024	CUST1531	Ahmad Halim	Mid-West	Beauty	Moisturizer	1	35.09	Completed	Gift Card	Beauty	
5	ORD101567	8/22/2024	CUST1390	Ayu Hidayat	SOUTH	Sports & Ou	Camping Tent	2	12.94	Completed	Debit Card	Sports & Outdoors	
6	ORD101096	21-Feb-24	CUST1563	James Yusuf	SOUTH	sports & ou	Hiking Backpack	1	23.12	Completed	PayPal	Sports & Outdoors	
7	ORD101473	8/18/2024	CUST1052	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	Beauty	
8	ORD103111	8/15/2024	CUST1726	John Tan	south	books	Fiction Novel	2	10.88	Completed	Gift Card	Books	
9	ORD100537	1/28/2024	CUST1445	Kevin Anderson	WEST	Beauty	Moisturizer	1	21.46	Completed	PayPal	Beauty	
10	ORD102413	24-Apr-23	CUST1699	John Yusuf	MIDWEST	Home & Kit	Cutting Board	1	102.66	Completed	PayPal	Home & Kitchen	
11	ORD103151	11-Aug-23	CUST1708	Ayu Smith	South	sports & ou	Resistance Bands	1	56.4	Completed	PayPal	Sports & Outdoors	
12	ORD102767	9/23/2023	CUST1868	Nina Lee	midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	Books	
13	ORD101868	15-Mar-24	CUST1211		south	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	Clothing	
14	ORD100749	7/27/2024	CUST1334	Fajar Santos	east	Home & Kit	Storage Container	1	112.68	Completed	Gift Card	Home & Kitchen	
15	ORD101857	8/5/2023	CUST1104	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	Electronics	
16	ORD100800	23-Jan-24	CUST1352	Lisa Taylor	MIDWEST	Beauty	Shampoo	2	23.18	Cancelled		Beauty	
17	ORD101695	20-May-24	CUST1255	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	Clothing	
18	ORD102093	1/25/2024	CUST1824	Linda Chen	east	Sports & Ou	Camping Tent	1		Cancelled	PayPal	Sports & Outdoors	
19	ORD101296	10/6/2024	CUST1003	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	Electronics	
20	ORD103842	9/17/2023	CUST1741	Siti Wilson	midwest	Sports & Ou	Yoga Mat	1	75.14	Completed	Credit Card	Sports & Outdoors	
21	ORD102931	10/26/2023	CUST1856		SOUTH	electronics	Power Bank	5	126.65	Completed	Gift Card	Electronics	
22	ORD102954	8/19/2023	CUST1210	Nina Garcia	West	Sports & Ou	Hiking Backpack	1	49.73	Returned	Credit Card	Sports & Outdoors	
23	ORD103824	7/22/2024	CUST1524	Robert Smith	west	clothing	Denim Jeans	4	62.83	Completed	Debit Card	Clothing	
24	ORD101074	4/12/2024	CUST1689	Siti Tan	EAST	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	Beauty	

Figure 4. The category_clean helper column using the PROPER(TRIM()) formula

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	region_trim	
1	ORD100016	2023-11-23	CUST1353	Andi Halim		Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card		
2	ORD102704	2023-03-14	CUST1478	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	East	
3	ORD103490	2024-04-10	CUST1531	Ahmad Halim	Midwest	Beauty	Moisturizer	1	35.09	Completed	Gift Card	Midwest	
4	ORD101567	2024-08-22	CUST1390	Ayu Hidayat	South	Sports & Outdoors	Camping Tent	2	12.94	Completed	Debit Card	South	
5	ORD101096	2024-02-21	CUST1563	James Yusuf	South	Sports & Outdoors	Hiking Backpack	1	23.12	Completed	PayPal	South	
6	ORD101473	2024-08-18	CUST1052	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	Midwest	
7	ORD103111	2024-08-15	CUST1726	John Tan	South	Books	Fiction Novel	2	10.88	Completed	Gift Card	South	
8	ORD100537	2024-01-28	CUST1445	Kevin Anderson	West	Beauty	Moisturizer	1	21.46	Completed	PayPal	West	
9	ORD102413	2023-04-24	CUST1699	John Yusuf	Midwest	Home & Kitchen	Cutting Board	1	102.66	Completed	PayPal	Midwest	
10	ORD103151	2023-08-11	CUST1708	Ayu Smith	South	Sports & Outdoors	Resistance Bands	1	56.4	Completed	PayPal	South	
11	ORD102767	2023-09-22	CUST1868	Nina Lee	Midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	Midwest	
12	ORD101868	2024-03-15	CUST1211		South	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	South	
13	ORD100749	2024-07-27	CUST1334	Fajar Santos	East	Home & Kitchen	Storage Container	1	112.68	Completed	Gift Card	East	
14	ORD101857	2024-08-05	CUST1104	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	East	
15	ORD100800	2024-01-23	CUST1352	Lisa Taylor	Midwest	Beauty	Shampoo	2	23.18	Cancelled		Midwest	
16	ORD101695	2024-05-20	CUST1255	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	West	
17	ORD102093	2024-01-25	CUST1824	Linda Chen	East	Sports & Outdoors	Camping Tent	1		Cancelled	PayPal	East	
18	ORD101296	2024-10-06	CUST1003	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	East	
19	ORD103842	2023-09-17	CUST1741	Siti Wilson	Midwest	Sports & Outdoors	Yoga Mat	1	75.14	Completed	Credit Card	Midwest	
20	ORD102931	2023-10-26	CUST1856		South	Electronics	Power Bank	5	126.65	Completed	Gift Card	South	
21	ORD102954	2023-08-19	CUST1210	Nina Garcia	West	Sports & Outdoors	Hiking Backpack	1	49.73	Returned	Credit Card	West	
22	ORD103824	2024-07-22	CUST1524	Robert Smith	West	Clothing	Denim Jeans	4	62.83	Completed	Debit Card	West	
23	ORD101074	2024-04-12	CUST1689	Siti Tan	East	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	East	
24	ORD101777	2024-12-28	CUST1227	David Wijaya	East	Home & Kitchen	Blender	4	115.54	Completed	Debit Card	East	
25	ORD102224	2023-05-05	CUST1588	David Anderson	South	Sports & Outdoors	Camping Tent	1	30.3	Completed	Debit Card	South	
26	ORD103815	2024-07-04	CUST1265	Sarah Johnson	South	Clothing	Hoodie	1	46.19	Completed	Debit Card	South	
27	ORD101819	2023-02-09	CUST1343	Dewi Sari	East	Books	Self-Help Guide	1	25.66	Completed	Credit Card	East	
28	ORD100756	2023-07-17	CUST1680	Putri Halim	East	Sports & Outdoors	Camping Tent	1	72.43	Cancelled	PayPal	East	
29	ORD101910	2023-04-08	CUST1887	Maria Wilson	South	Toys	Board Game	1	58.15	Completed	PayPal	South	
30	ORD100326	2024-07-07	CUST1182	Dewi Tan	East	Beauty	Lip Balm	2	7.83	Completed	PayPal	East	
31	ORD103238	2023-04-16	CUST1254	Nina Garcia	Midwest	Toys	RC Car	1	9.57	Completed		Midwest	
32	ORD102634	2023-05-24	CUST1757	Sarah Lee	East	Clothing	Cotton T-Shirt	1	54.55	Completed	Debit Card	East	
33	ORD103616	2024-02-26	CUST1128	Rina Taylor	South	Beauty	Moisturizer	1	25.47	Completed	Debit Card	South	
34	ORD101939	2023-11-15	CUST1806	Ahmad Hidayat	Midwest	Home & Kitchen	Coffee Maker	2	52.95	Cancelled	Gift Card	Midwest	
35	ORD103226	2024-09-07	CUST1206	Siti Wilson	West	Beauty	Shampoo	2	\$44.80	Completed	PayPal	West	
36	ORD102507	2023-09-23	CUST1645	Sarah Chen	West	Toys	Board Game	1	58.26	Completed	Debit Card	West	
37	ORD100857	2023-01-17	CUST1370	Rina Anderson	West	Toys	Puzzle 1000pc	1	25.68	Completed	Debit Card	West	
38	ORD102423	2023-05-01	CUST1838	Budi Chen	South	Electronics	Bluetooth Speaker	2	93.44	Completed	PayPal	South	
39	ORD100333	2024-09-25	CUST1753	Kevin Lee	East	Clothing	Running Shorts	2	24.84	Completed	PayPal	East	
40	ORD102984	2023-09-17	CUST1615	Putri Anderson	West	Clothing	Hoodie	2		Completed	PayPal	West	
41	ORD100087	2023-03-06	CUST1867	Nina Garcia	South	Beauty	Moisturizer	2	10.48	Completed	Gift Card	South	
42	ORD100619	2024-06-21	CUST1414	Ayu Chen	South	Electronics	Power Bank	2	142.03	Completed	Debit Card	South	

Figure 5. The category and region columns after standardization (Proper Case, no typos)

3.3 Standardizing the Date Format (order_date)

The order_date column initially contained 5 different formats (YYYY-MM-DD, MM/DD/YYYY, DD-Mon-YYYY, DD/MM/YYYY, YYYY/MM/DD). Rows still stored as text were first identified using ISTEXT:

```
=ISTEXT([@order_date])
```

	id	is_text_date
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
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		FALSE
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		FALSE
		FALSE
		TRUE
		FALSE
		FALSE

Figure 6. The `is_text_date` helper column — `TRUE` marks rows whose `order_date` is still stored as text

Rows flagged TRUE were converted using the formula below, which explicitly reads the DD/MM/YYYY format and leaves already-valid dates (FALSE) unchanged:

```
=IF(ISTEXT([@order_date]), DATE(VALUE(RIGHT([@order_date],4)),  
VALUE(MID([@order_date],4,2)), VALUE(LEFT([@order_date],2))), [@order_date])
```

Once every value was successfully converted into a real date, the display format was standardized via Format Cells > Date > custom type YYYY-MM-DD (rather than the asterisked format that follows Windows regional settings):

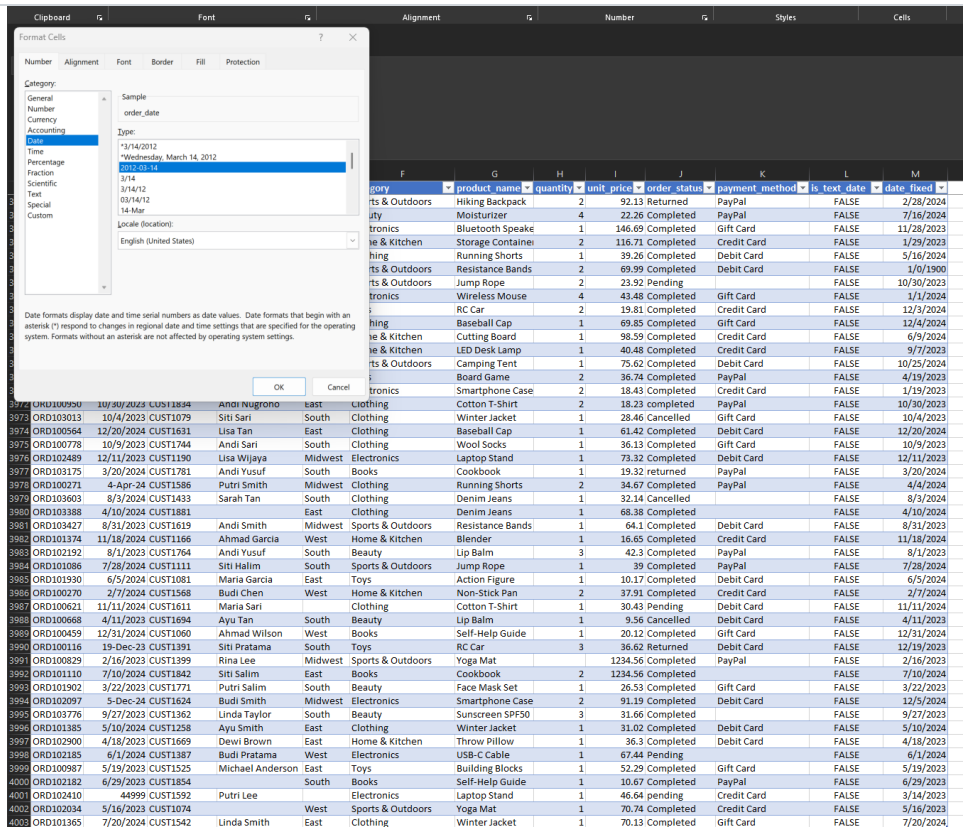


Figure 7. The Format Cells dialog — choosing the YYYY-MM-DD date format that stays consistent across systems

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	is_text_date	date
1	ORD100016	2023-11-23	CUST1353	Andi Halim	East	Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card	FALSE	2/28/2024
2	ORD102704	2023-03-14	CUST1478	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	FALSE	7/16/2024
3	ORD103490	2024-04-10	CUST1531	Ahmad Halim	Midwest	Beauty	Moisturizer	1	35.09	Completed	Gift Card	FALSE	11/28/2023
4	ORD101567	2024-08-22	CUST1390	Ayu Hidayat	SOUTH	Sports & Outdoors	Camping Tent	2	12.94	Completed	Debit Card	FALSE	1/29/2023
5	ORD101096	2024-02-21	CUST1563	James Yusuf	SOUTH	Sports & Outdoors	Hiking Backpack	1	23.12	Completed	PayPal	FALSE	5/16/2024
6	ORD101473	2024-08-18	CUST1052	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	FALSE	1/0/1900
7	ORD103111	2024-08-15	CUST1726	John Tan	South	Books	Fiction Novel	2	10.88	Completed	Gift Card	FALSE	10/30/2023
8	ORD100537	2024-01-28	CUST1445	Kevin Anderson	WEST	Beauty	Moisturizer	1	21.46	Completed	PayPal	FALSE	1/1/2024
9	ORD102413	2023-04-24	CUST1699	John Yusuf	MIDWEST	Home & Kitchen	Cutting Board	1	102.66	Completed	PayPal	FALSE	12/3/2024
10	ORD103151	2023-08-11	CUST1708	Ayu Smith	South	Sports & Outdoors	Resistance Bands	1	56.4	Completed	PayPal	FALSE	12/4/2024
11	ORD102767	2023-09-22	CUST1868	Nina Lee	Midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	FALSE	6/9/2024
12	ORD101868	2024-03-15	CUST1211		South	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	FALSE	9/7/2023
13	ORD100749	2024-07-27	CUST1334	Fajar Santos	East	Home & Kitchen	Storage Container	1	112.68	Completed	Gift Card	FALSE	10/25/2024
14	ORD101857	2023-08-05	CUST1104	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	FALSE	4/19/2023
15	ORD100800	2024-01-23	CUST1352	Lisa Taylor	MIDWEST	Beauty	Shampoo	2	23.18	Cancelled		FALSE	4/19/2023
16	ORD101695	2024-05-20	CUST1255	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	FALSE	8/3/2024
17	ORD102093	2024-01-25	CUST1824	Linda Chen	East	Sports & Outdoors	Camping Tent	1		Cancelled	PayPal	FALSE	4/1/2023
18	ORD101296	2024-10-06	CUST1003	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	FALSE	12/31/2024
19	ORD103842	2023-09-17	CUST1741	Siti Wilson	Midwest	Sports & Outdoors	Yoga Mat	1	75.14	Completed	Credit Card	FALSE	10/4/2023
20	ORD102931	2023-10-26	CUST1856		SOUTH	Electronics	Power Bank	5	126.65	Completed	Gift Card	FALSE	3/2/2023
21	ORD102954	2023-08-19	CUST1210	Nina Garcia	West	Sports & Outdoors	Hiking Backpack	1	49.73	Returned	Credit Card	FALSE	12/19/2023
22	ORD103824	2024-07-22	CUST1524	Robert Smith	West	Clothing	Denim Jeans	4	62.83	Completed	Debit Card	FALSE	10/9/2023
23	ORD101074	2024-04-12	CUST1689	Siti Tan	EAST	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	FALSE	12/12/2024
24	ORD101777	2024-12-28	CUST1227	David Wijaya	EAST	Home & Kitchen	Blender	4	115.54	Completed	Debit Card	FALSE	3/20/2024
25	ORD102224	2023-05-05	CUST1588	David Anderson	South	Sports & Outdoors	Camping Tent	1	30.3	Completed	Debit Card	FALSE	4/4/2024
26	ORD103815	2024-07-04	CUST1265	Sarah Johnson	South	Clothing	Hoodie	1	46.19	Completed	Debit Card	FALSE	8/3/2024
27	ORD101819	2023-02-09	CUST1343	Dewi Sari	EAST	Books	Self-Help Guide	1	25.66	Completed	Credit Card	FALSE	4/10/2024
28	ORD100756	2023-07-17	CUST1680	Putri Halim	East	Sports & Outdoors	Camping Tent	1	72.43	Cancelled	PayPal	FALSE	8/1/2023
29	ORD101910	2023-04-08	CUST1887	Maria Wilson	SOUTH	Toys	Board Game	1	58.15	Completed	PayPal	FALSE	11/18/2024
30	ORD100326	2024-07-07	CUST1182	Dewi Tan	East	Beauty	Lip Balm	2	7.83	Completed	PayPal	FALSE	8/1/2023
31	ORD103238	2023-04-16	CUST1254	Nina Garcia	MIDWEST	Toys	RC Car	1	9.57	Completed		FALSE	7/28/2024
32	ORD102634	2023-05-24	CUST1757	Sarah Lee	East	Clothing	Cotton T-Shirt	1	54.55	Completed	Debit Card	FALSE	10/25/2024
33	ORD103616	2024-02-26	CUST1128	Rina Taylor	SOUTH	Beauty	Moisturizer	1	25.47	Completed	Debit Card	FALSE	10/30/2023
34	ORD101939	2023-11-15	CUST1806	Ahmad Hidayat	Midwest	Home & Kitchen	Coffee Maker	2	52.95	Cancelled	Gift Card	FALSE	10/4/2023
35	ORD103226	2024-09-07	CUST1206	Siti Wilson	WEST	Beauty	Shampoo	2	\$44.80	Completed	PayPal	FALSE	9/7/2023
36	ORD102507	2023-09-23	CUST1645	Sarah Chen	WEST	Toys	Board Game	1	58.26	Completed	Debit Card	FALSE	10/21/2023
37	ORD100857	2023-01-17	CUST1370	Rina Anderson	WEST	Toys	Puzzle 1000pc	1	25.68	Completed	Debit Card	FALSE	12/19/2023
38	ORD102423	2023-05-01	CUST1838	Budi Chen	South	Electronics	Bluetooth Speaker	2	93.44	Completed	PayPal	FALSE	12/3/2024
39	ORD100333	2024-09-25	CUST1753	Kevin Lee	East	Clothing	Running Shorts	2	24.84	Completed	PayPal	FALSE	12/4/2024
40	ORD102984	2023-09-17	CUST1615	Putri Anderson	West	Clothing	Hoodie	2		Completed	PayPal	FALSE	6/9/2024
41	ORD100087	2023-03-06	CUST1867	Nina Garcia	SOUTH	Beauty	Moisturizer	2	10.48	Completed	Gift Card	FALSE	9/7/2023
42	ORD100619	2024-06-21	CUST1414	Ayu Chen	South	Electronics	Power Bank	2	142.03	Completed	Debit Card	FALSE	10/25/2024
43												FALSE	4/19/2023

Figure 8. The order_date column after every value was converted and uniformly formatted

- The entire order_date column was reformatted uniformly to YYYY-MM-DD.
- One date value stored as 0 (a former blank cell) was found via =MIN(order_date) and reverted to blank so it would not disguise a missing value as a 1900 date.

	A	B	C	D	E	F
1	Hari	Fase	Task	Tools	Selesai?	
2	Day 1	Excel Data Cleaning	Ambil dataset e-commerce sales	Excel	SELESAI	
3	Day 1	Excel Data Cleaning	Cek & tangani missing values	Excel		
4	Day 1	Excel Data Cleaning	Cek & hapus duplicate records	Excel	SELESAI	
5	Day 1	Excel Data Cleaning	Perbaiki data types yang salah	Excel		
6	Day 1	Excel Data Cleaning	Standarkan kategori yang tidak konsisten	Excel	SELESAI	
7	Day 1	Excel Data Cleaning	Format ulang kolom tanggal	Excel	SELESAI	
8	Day 1	Excel Data Cleaning	Cek outlier	Excel		
9	Day 1	Excel Data Cleaning	Buat calculated columns	Excel		
10	Day 1	Excel Data Cleaning	Dokumentasikan raw -> cleaning -> clean dataset	Excel		
11	Day 1	Excel Data Cleaning	Simpan sebagai cleaned sales data.xlsx	Excel		

Figure 9. Day 1 checklist after the duplicate, data types, category/region, and date steps are complete

3.4 Fixing Data Types and Outliers (Quantity & Unit Price)

The quantity and unit_price columns had several problems at once: text-typed values ("one", "two"), negative values, prices stored as text in "\$xx.xx" format, and extreme outliers implausible for individual retail transactions.

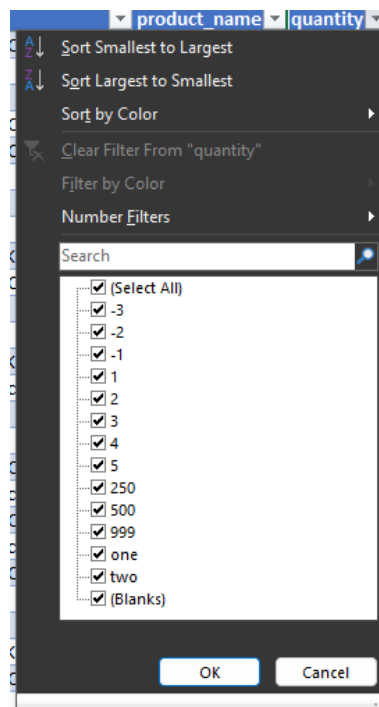


Figure 10. Problematic values in the quantity column, seen via the filter dropdown (negative, text, and extreme outliers all mixed together)

G	H	I	J	K	L
product_name	quantity	unit_price	order_status	payment_method	quantity_clean
Wool Socks	-3	64.63	Completed	PayPal	3
Fiction Novel	-3	30.39	Cancelled	Credit Card	3

Figure 11. Example rows with a negative quantity (-3) before being fixed

Fixing the quantity data type

```
quantity_clean:
=IFERROR(IF([@quantity]="one",1,IF([@quantity]="two",2,IF([@quantity]="three",3,ABS(VALUE([@quantity]))))),"
```

Fixing the unit_price data type

	H	I	J	K	L
	quantity	unit_price	order_status	payment_method	quantity_clean
	2	\$44.80	Completed	PayPal	2
	1	\$75.09	Completed	PayPal	1
	1	\$86.87	Completed	PayPal	1
	1	\$69.18	Completed	Credit Card	1
	1	\$17.17	Completed	Debit Card	1
	1	\$60.06	Completed	Gift Card	1
	1	\$20.53	Completed	PayPal	1
	1	\$31.32	Completed	PayPal	1
	1	\$106.64	Completed	Debit Card	1
	2	\$34.82	Completed	Credit Card	2
	1	\$41.30	Completed	Gift Card	1
	1	\$130.75	Completed	Credit Card	1
	1	\$29.64	Completed		1
	1	\$16.59	completed	Debit Card	1
	1	\$38.05	Completed	Gift Card	1
	1	\$44.88	Completed	Credit Card	1
	2	\$12.32	Completed	Debit Card	2
	2	\$75.42	Completed	Debit Card	2
	3	\$15.48	cancelled	Gift Card	3

Figure 12. The unit_price column still in "\$xx.xx" text format before cleanup

```
unit_price_clean:
=IFERROR(VALUE(SUBSTITUTE([@unit_price],"$","")), "")
```

	H	I	J	K	L	M
	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price
	2	\$44.80	Completed	PayPal	2	44.8
	1	\$75.09	Completed	PayPal	1	75.09
	1	\$86.87	Completed	PayPal	1	86.87
	1	\$69.18	Completed	Credit Card	1	69.18
	1	\$17.17	Completed	Debit Card	1	17.17
	1	\$60.06	Completed	Gift Card	1	60.06
	1	\$20.53	Completed	PayPal	1	20.53
	1	\$31.32	Completed	PayPal	1	31.32
	1	\$106.64	Completed	Debit Card	1	106.64
	2	\$34.82	Completed	Credit Card	2	34.82
	1	\$41.30	Completed	Gift Card	1	41.3
	1	\$130.75	Completed	Credit Card	1	130.75
	1	\$29.64	Completed		1	29.64
	1	\$16.59	completed	Debit Card	1	16.59
	1	\$38.05	Completed	Gift Card	1	38.05
	1	\$44.88	Completed	Credit Card	1	44.88
	2	\$12.32	Completed	Debit Card	2	12.32
	2	\$75.42	Completed	Debit Card	2	75.42
	3	\$15.48	cancelled	Gift Card	3	15.48

Figure 13. The unit_price_clean column — the \$ sign successfully removed, leaving pure numbers

Identifying and removing outliers

Once the data types were clean, outliers were reviewed by sorting (Sort Largest to Smallest) and examining the context of each row (product, customer, category):

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price	
1	ORD100335	2023-08-07	CUST10411	Ahmad Garcia	West	Home & Kitchen	Coffee Maker	999	43.04	Completed	PayPal	999	43.04	
2	ORD101178	2023-08-01	CUST1004	Ahmad Yusuf	South	Toys	Action Figure	999	35.43	completed	Gift Card	999	35.43	
3	ORD102638	2023-02-23	CUST10083	Ahmad Taylor	East	Sports & Outdoors	Yoga Mat	999	71.18	Completed	Gift Card	999	71.18	
4	ORD103218	2023-10-28	CUST1474	Kevin Sari	East	Beauty	Sunscreen SPF50	999	9.1	Completed	Gift Card	999	9.1	
5	ORD102875	2023-04-30	CUST1808	Michael Kurniawan	Midwest	Sports & Outdoors	Jump Rope	999	85.82	Completed	Credit Card	999	85.82	
6	ORD100684	2023-09-16	CUST1652	Ahmad Smith	West	Books	Business Strategy	999	17.71	Completed	Debit Card	999	17.71	
7	ORD102343	2023-07-25	CUST1772	Ahmad Hidayat		Electronics	Wireless Earbuds	500	52.43	Returned	Gift Card	500	52.43	
8	ORD103588	2023-01-01	CUST1104	Sarah Tan	South	Sports & Outdoors	Water Bottle	500	32.5	Cancelled	PayPal	500	32.5	
9	ORD100867	2023-01-31	CUST1399	Dewi Chen	West	Clothing	Winter Jacket	500	32.44	Completed	Gift Card	500	32.44	
10	ORD102002	2023-05-27	CUST1262	Robert Garcia	Midwest	Clothing	Hoodie	500	13.13	Completed	Credit Card	500	13.13	
11	ORD101689	2023-07-10	CUST1290	Lisa Nugroho	West	Home & Kitchen	Non-Stick Pan	500	98.01	Completed	Debit Card	500	98.01	
12	ORD100031	2024-05-22	CUST1179	David Salim	East	Sports & Outdoors	Water Bottle	250	9.42	Completed	Debit Card	250	9.42	
13	ORD100209	2024-01-20	CUST1761	Ahmad Johnson	East	Toys	Building Blocks	250	53.78	Completed		250	53.78	
14	ORD103734	2024-12-01	CUST1871	Andi Smith	West	Electronics	Laptop Stand	250	127.77	Cancelled	Debit Card	250	127.77	
15	ORD102727	2023-09-04	CUST1713	Dewi Taylor	West	Electronics	Webcam	250	44.7	Completed	Debit Card	250	44.7	
16	ORD102931	2023-10-26	CUST1856		South	Electronics	Power Bank	5	126.65	Completed	Gift Card	5	126.65	
17	ORD103343	2023-12-09	CUST1383	David Garcia	West	Sports & Outdoors	Hiking Backpack	5	68.67	Completed	Debit Card	5	68.67	
18	ORD100234	2023-12-22	CUST1766	Michael Kurniawan	South	Beauty	Sunscreen SPF50	5	37.89	Completed	PayPal	5	37.89	
19	ORD102815	2023-01-17	CUST1555	Andi Hidayat	Midwest	Toys	Action Figure	5	48.39	Completed	Debit Card	5	48.39	

Figure 14. The quantity_clean column sorted from largest — 15 rows with values of 999/500/250 are visible

19	ORD100533	2024-09-25	CUST1716	Budi Salim	Midwest	Toys	Puzzle 1000pc	5	8888	Completed	Gift Card	5	8888	
20	ORD101663	2024-08-01	CUST1572	Maria Yusuf	East	Toys	Puzzle 1000pc	5	51.6	Cancelled	Debit Card	5	51.6	

Figure 15. One of the quantity-outlier rows (Puzzle 1000pc) also happened to have an extreme unit_price (8888)

- Quantity: 15 rows with values of 250/500/999 were removed. The values were scattered randomly across various products and customers with no wholesale-order pattern — classified as a data entry error, and immaterial at only 0.37% of the dataset.

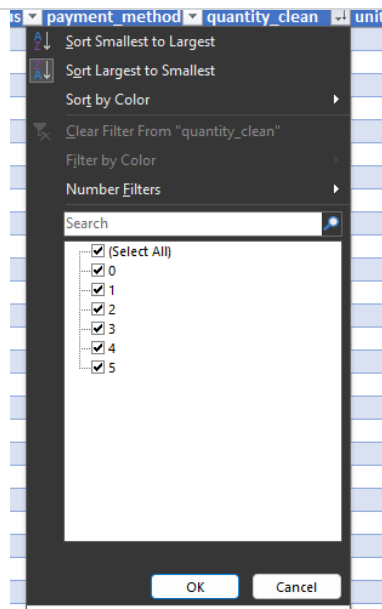


Figure 16. The quantity column filter after 15 outlier rows were removed — only values 0–5 remain

77	ORD101914	2023-09-23	CUST1838	Budi Chen	South	Home & Kitchen	Throw Pillow			Completed	PayPal		0
78	ORD100143	2024-10-24	CUST1799	Robert Hidayat	South	Toys	Building Blocks			Completed	Credit Card		0
79	ORD100533	2024-09-25	CUST1716	Budi Salim	Midwest	Toys	Puzzle 1000pc	5	8888	Completed	Gift Card	5	8888
80	ORD100693	2024-03-23	CUST1630	John Kurniawan	West	Home & Kitchen	Blender	1	8888	Returned	PayPal	1	8888
81	ORD101843	2024-12-16	CUST1023	Siti Wilson	East	Home & Kitchen	Non-Stick Pan	1	8888	Completed	Credit Card	1	8888
82	ORD102699	2024-01-20	CUST1454	Fajar Wijaya	Midwest	Clothing	Hoodie	1	8888	Completed	Gift Card	1	8888
83	ORD102755	2023-11-10	CUST1809	Kevin Smith	Midwest	Books	Business Strategy	2	4999.99	Completed	PayPal	2	4999.99
84	ORD101880	2024-05-06	CUST1535	Maria Brown	Midwest	Toys	Action Figure	1	4999.99	Cancelled	Credit Card	1	4999.99
85	ORD100921	2024-05-15	CUST1725	Nina Pratama	West	Books	Business Strategy	2	1234.56	Completed	Credit Card	2	1234.56
86	ORD101822	2024-11-14	CUST1564	Maria Chen	South	Sports & Outdoors	Jump Rope	2	1234.56	Completed	Credit Card	2	1234.56
87	ORD101110	2024-07-10	CUST1842	Siti Salim	East	Books	Cookbook	2	1234.56	Completed		2	1234.56
88	ORD101599	2024-12-06	CUST1138	Fajar Anderson	Midwest	Electronics	Power Bank	1	1234.56	Completed	Gift Card	1	1234.56
89	ORD101660	2024-01-01	CUST1145	Linda Smith	Midwest	Clothing	Wool Socks	1	1234.56	Completed	Debit Card	1	1234.56
90	ORD100829	2023-02-16	CUST1399	Rina Lee	Midwest	Sports & Outdoors	Yoga Mat	1	1234.56	Completed	PayPal	0	1234.56
91	ORD101912	2024-07-21	CUST1634	James Sari	East	Electronics	Power Bank	2	149.71	Pending	Debit Card	2	149.71

Figure 17. 12 rows with extreme unit_price values (8888 / 4999.99 / 1234.56) selected for removal

- Unit price: 12 rows with values of 8888 / 4999.99 / 1234.56 were removed. These values did not match their product category (e.g., a Puzzle 1000pc priced at \$8,888), and the value 1234.56 appeared identically across 6 different products — strong evidence of a system error rather than legitimate price variation.

T	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order ID	Order Date	Customer Name	Address	City	State	Product Name	Quantity	Unit Price	Order Status	Payment Method	Quantity Clean	Unit Price	
3933	ORD1001018	2023-02-11	CUST1260	Fajar Pratama	East	Sports & Outdoors	Resistance Bands	1		Completed	Gift Card	1		
3933	ORD100917	2024-03-21	CUST1194	Maria Hidayat	South	Sports & Outdoors	Hiking Backpack	1		Returned	PayPal	1		
3933	ORD100944	2024-05-22	CUST1762	Rina Nugroho	South	Sports & Outdoors	Resistance Bands	1		Completed	Credit Card	1		
3933	ORD100984	2024-05-16	CUST1380	Lisa Brown	West	Beauty	Sunscreen SPF50	1		Completed	PayPal	1		
3933	ORD101530	2024-06-30	CUST1287	John Pratama	South	Sports & Outdoors	Yoga Mat	1		Pending	PayPal	1		
3933	ORD101879	2024-04-05	CUST1136	Budi Hidayat	Midwest	Books	Self-Help Guide	1		Completed	Debit Card	1		
3940	ORD103237	2023-10-11	CUST1624	Budi Smith	South	Home & Kitchen	Cutting Board	1		Completed	Gift Card	1		
3940	ORD100905	2024-05-15	CUST1350	Linda Salim	Midwest	Clothing	Hoodie	1		Completed	Credit Card	1		
3940	ORD103913	2023-01-24	CUST1606	Putri Yusuf	Midwest	Home & Kitchen	Cutting Board	1		Completed	Credit Card	1		
3940	ORD101125	2023-01-27	CUST1516	Fajar Brown	Midwest	Electronics	Webcam	1		Returned	Gift Card	1		
3940	ORD103970	2023-04-26	CUST1329	Andi Tan	South	Clothing	Denim Jeans	1		Returned	PayPal	1		
3940	ORD101487	2023-08-21	CUST1687	Siti Sari	South	Toys	Action Figure	1		Completed	Credit Card	1		
3940	ORD103406	2024-08-14	CUST1349	John Hallim	East	Toys	Board Game	1		Completed	Gift Card	1		
3940	ORD102133	2024-11-30	CUST1745	Ahmad Hallim	East	Clothing	Denim Jeans	1		Completed	Debit Card	1		
3940	ORD102818	2023-08-09	CUST1660	Andi Salim	East	Sports & Outdoors	Hiking Backpack	1		Completed	Credit Card	1		
3940	ORD100019	2024-08-06	CUST1412	Siti Anderson	South	Home & Kitchen	Coffee Maker	1		Completed	Debit Card	1		
3950	ORD101245	2024-08-24	CUST1412	Siti Anderson	West	Clothing	Wool Socks	1		Cancelled	PayPal	1		
3950	ORD103533	2024-02-27	CUST1894	Putri Anderson	East	Home & Kitchen	Blender	1		Completed	Gift Card	1		
3950	ORD101540	2024-06-01	CUST1754	Kevin Salim	West	Electronics	Power Bank	1		Returned	Debit Card	1		
3950	ORD100027	2023-05-05	CUST1324	Budi Santos	Midwest	Sports & Outdoors	Resistance Bands	1		Completed	Gift Card	1		
3950	ORD102052	2024-11-04	CUST1837	Devi Johnson	Midwest	Home & Kitchen	Blender	1		Completed	Debit Card	1		
3950	ORD103958	2024-01-30	CUST1005	Oswi Wijaya	Midwest	Toys	Building Blocks	1		Completed	Gift Card	1		
3950	ORD101555	2023-08-18	CUST1168	Linda Nugroho	East	Toys	Board Game	1		Completed	PayPal	1		
3950	ORD103247	2024-11-26	CUST1177	Ayu Pratama	East	Beauty	Sunscreen SPF50	1		Completed	Debit Card	1		
3950	ORD102778	2023-04-08	CUST1161	Ayu Yusuf	East	Books	Business Strategy	1		Completed	Credit Card	1		
3950	ORD101476	2024-01-01	CUST1171	James Anderson	Midwest	Books	Children's Book	1		Completed	PayPal	1		
3960	ORD100166	2024-10-20	CUST1846	Kevin Kurniawan	East	Books	Business Strategy	1		Returned	Gift Card	1		
3960	ORD100067	2023-07-01	CUST1617	Robert Tan	South	Beauty	Shampoo	1		Pending	Debit Card	1		
3960	ORD101989	2023-08-31	CUST1684	Fajar Salim	East	Toys	Puzzle 1000pc	1		Completed	PayPal	1		
3960	ORD103234	2024-09-18	CUST1642	John Hallim	South	Clothing	Hoodie	1		Returned	Gift Card	1		
3960	ORD103178	2024-11-30	CUST1256	James Wijaya	West	Clothing	Winter Jacket	1		Completed	Credit Card	1		
3960	ORD101526	2023-10-22	CUST1811	Putri Santos	Midwest	Clothing	Denim Jeans	1		Completed	PayPal	1		
3960	ORD100898	2023-05-02	CUST1575		West	Clothing	Denim Jeans	one		Completed	Gift Card	1		
3960	ORD103206	2024-05-17	CUST1417		Electronics	Webcam		1		Completed	PayPal	1		
3960	ORD103339	2024-11-25	CUST1554	Michael Tan	East	Electronics	Laptop Stand	1		Completed	PayPal	1		
3960	ORD100759	2023-11-14	CUST1252	Ayu Brown	West	Home & Kitchen	Throw Pillow	1		Completed	Debit Card	1		
3970	ORD103614	2024-08-27	CUST1386	Lisa Johnson	Midwest	Books	Business Strategy	1		Completed	Credit Card	1		
3970	ORD102400	2024-07-10	CUST1324	Budi Santos	Midwest	Toys	Action Figure	1		Completed	Debit Card	1		
3970	ORD101394	2024-07-13	CUST1765	Ayu Brown	Midwest	Electronics	Smartphone Case	1		Completed	Credit Card	0		
3970	ORD100216	2024-09-12	CUST1356	Ahmad Tan	Midwest	Home & Kitchen	Throw Pillow	1		Pending	Gift Card	0		
3970	ORD101914	2023-09-23	CUST1838	Budi Chen	South	Home & Kitchen	Throw Pillow	1		Completed	PayPal	0		
3970	ORD100143	2024-10-24	CUST1799	Robert Hidayat	South	Toys	Building Blocks	1		Completed	Credit Card	0		

Figure 18. Before: quantity/unit_price outliers were still mixed into the data

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price_clean
7	ORD100006	2024-01-16	CUST1580	Linda Tan	South	Electronics	Smartphone Case	1	81.85	Completed	Credit Card	1	81.85
8	ORD100007	2023-09-26	CUST1753	Kevin Lee	South	Clothing	Running Shorts	1	77.87	Completed	Gift Card	1	77.87
9	ORD100008	2024-11-29	CUST1075	Linda Anderson	South	Sports & Outdoors	Water Bottle	1	43.76	Pending	Debit Card	1	43.76
10	ORD100009	2023-05-05	CUST1726	John Tan	Midwest	Beauty	Shampoo	1	36.12	Pending	PayPal	1	36.12
11	ORD100010	2024-02-12	CUST1332	Robert Tan	East	Books	Fiction Novel	1	10.00	Pending	Gift Card	1	10
12	ORD100011	2024-07-18	CUST1460	Rina Sari	Midwest	Sports & Outdoors	Jump Rope	3	36.61	Completed	PayPal	3	36.61
13	ORD100012	2023-12-21	CUST1311	John Nugroho	Clothing	Denim jeans	3	18.37	Returned	Debit Card	3	18.37	
14	ORD100013	2024-11-05	CUST1546	John Garcia	West	Electronics	Smartphone Case	1	72.72	Completed	Debit Card	1	72.72
15	ORD100014	1900-01-00	CUST1590	Putri Tan	West	Home & Kitchen	LED Desk Lamp	3	93.60	Completed	Gift Card	3	93.6
16	ORD100015	2023-06-11	CUST1515	Sarah Salim	East	Home & Kitchen	Cutting Board	4	106.12	Completed	Gift Card	4	106.12
17	ORD100016	2023-11-23	CUST1353	Andi Halim	Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card	2	28.97	
18	ORD100017	2023-07-14	CUST1254	Nina Garcia	Midwest	Toys	Puzzle 1000pc	1	25.48	Completed	Gift Card	1	25.48
19	ORD100018	2023-02-21	CUST1468	Michael Tan	Midwest	Clothing	Denim jeans	1	35.43	Completed	Gift Card	1	35.43
20	ORD100019	2024-08-06	CUST1412	Siti Anderson	South	Home & Kitchen	Coffee Maker	1		Completed	Debit Card	1	
21	ORD100020	2024-02-08	CUST1696	David Hidayat	West	Books	Fiction Novel	1	28.56	Completed	Debit Card	1	28.56
22	ORD100021	2024-06-15	CUST1039	Robert Tan	Midwest	Clothing	Wool Socks	1	54.35	Completed	PayPal	1	54.35
23	ORD100022	2024-01-05	CUST1344	Ahmad Salim	South	Books	Business Strategy	1	14.12	Completed	Debit Card	1	14.12
24	ORD100023	2023-02-22	CUST1544	Andi Salim	Midwest	Electronics	Wireless Mouse	2	67.21	Completed	PayPal	2	67.21
25	ORD100024	2024-03-27	CUST1586	Putri Smith	West	Beauty	Moisturizer	1	9.63	Completed	PayPal	1	9.63
26	ORD100025	2024-11-09	CUST1744	Andi Sari	West	Electronics	Smartphone Case	3	100.47	Completed	Gift Card	3	100.47
27	ORD100026	2023-09-10	CUST1738	Fajar Halim	South	Beauty	Face Mask Set	2	9.96	Completed	Debit Card	2	9.96
28	ORD100027	2023-05-05	CUST1324	Budi Santos	Midwest	Sports & Outdoors	Resistance Bands	1		Completed	Gift Card	1	
29	ORD100028	2024-05-05	CUST1731	Robert Anderson	South	Electronics	Laptop Stand	1	115.78	Completed	Debit Card	1	115.78
30	ORD100029	2023-04-13	CUST1585	Budi Kurniawan	West	Books	Cookbook	1	25.04	Completed	Gift Card	1	25.04
31	ORD100030	2023-10-22	CUST1416		South	Beauty	Lip Balm	1	31.17	Completed	Credit Card	1	31.17
32	ORD100032	2024-05-31	CUST1656	Ayu Yusuf	South	Books	Business Strategy	4	16.56	Completed		4	16.56
33	ORD100033	2024-09-17	CUST1345	Michael Santos	West	Toys	RC Car	4	35.43	Cancelled	Gift Card	4	35.43
34	ORD100034	2023-08-30	CUST1331	Ayu Chen	Midwest	Clothing	Cotton T-Shirt	3	58.79	Completed	Debit Card	3	58.79
35	ORD100035	2023-06-13	CUST1063	David Anderson	South	Beauty	Moisturizer	3	33.38	Completed	Debit Card	3	33.38
36	ORD100036	2024-12-19	CUST1489	Nina Salim	Midwest	Clothing	Running Shorts	3	36.29	Completed	Gift Card	3	36.29
37	ORD100037	1900-01-00	CUST1806	Ahmad Hidayat	Midwest	Electronics	Wireless Mouse	1	94.76	Completed	PayPal	1	94.76
38	ORD100038	2024-04-09	CUST1597	Ahmad Nugroho	South	Sports & Outdoors	Yoga Mat	2	12.75	Completed	Gift Card	2	12.75
39	ORD100039	2024-01-09	CUST1836	Kevin Salim	East	Electronics	USB-C Cable	2	146.74	Completed	Credit Card	2	146.74
40	ORD100040	2024-01-27	CUST1184	Michael Kurniawan	South	Toys	Puzzle 1000pc	1	21.76	Completed	Gift Card	1	21.76
41	ORD100041	2023-01-04	CUST1182	Dewi Tan	West	Books	Children's Book	0	27.92	Completed	Debit Card	0	27.92
42	ORD100042	2024-07-11	CUST1158	Nina Nugroho	East	Home & Kitchen	Non-Stick Pan	1	93.85	Completed	Credit Card	1	93.85
43	ORD100043	2023-08-28	CUST1250	Linda Smith	South	Beauty	Moisturizer	3	12.75	Completed	Credit Card	3	12.75
44	ORD100044	2023-05-27	CUST1391	Fajar Wilson	West	Books	Self-Help Guide	1	24.90	Pending	PayPal	1	24.9
45	ORD100045	2023-12-19	CUST1695	James Johnson	South	Sports & Outdoors	Hiking Backpack	2	10.44	Returned	Gift Card	2	10.44
46	ORD100046	2024-12-25	CUST1484	James Anderson	West	Clothing	Baseball Cap	1	42.08	Completed	Gift Card	1	42.08

Figure 19. After: the quantity and unit_price columns are clean, with no extreme values

Hari	Fase	Task	Tools	Selesai?
Day 1	Excel Data Cleaning	Ambil dataset e-commerce sales	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek & tangani missing values	Excel	
Day 1	Excel Data Cleaning	Cek & hapus duplicate records	Excel	SELESAI
Day 1	Excel Data Cleaning	Perbaiki data types yang salah	Excel	SELESAI
Day 1	Excel Data Cleaning	Standarkan kategori yang tidak konsisten	Excel	SELESAI
Day 1	Excel Data Cleaning	Format ulang kolom tanggal	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek outlier	Excel	SELESAI
Day 1	Excel Data Cleaning	Buat calculated columns	Excel	
Day 1	Excel Data Cleaning	Dokumentasikan raw -> cleaning -> clean dataset	Excel	
Day 1	Excel Data Cleaning	Simpan sebagai cleaned_sales_data.xlsx	Excel	

Figure 20. Day 1 checklist after the data-type-fixing and outlier steps are complete

3.5 Handling Missing Values

After the previous steps, six columns still had missing values, differing in count and criticality. The number of missing values per column was calculated using COUNTBLANK in an empty area outside the table:

```
=COUNTBLANK(D2:D3975) {replace the column letter with the field being checked}
```

SUM	:	X	✓	f _x	=COUNTBLANK(D2:D3975)
1	order_id	order_date	customer_id	customer_name	region
3963	ORD103988	2024-04-15	CUST1412	Siti Anderson	West
3964	ORD103989	2023-06-18	CUST1696	David Hidayat	West
3965	ORD103990	2024-12-01	CUST1774	Maria Wijaya	West
3966	ORD103991	2023-11-10	CUST1664	Dewi Nugroho	Midwest
3967	ORD103992	2023-04-25	CUST1292	Budi Halim	South
3968	ORD103993	2024-03-03	CUST1382	David Hidayat	Midwest
3969	ORD103994	2024-01-17	CUST1058	John Johnson	West
3970	ORD103995	2023-02-16	CUST1113	Dewi Nugroho	Midwest
3971	ORD103996	2023-02-05	CUST1248	Andi Pratama	East
3972	ORD103997	2024-02-22	CUST1494	Robert Pratama	East
3973	ORD103998	2023-11-05	CUST1348	Putri Nugroho	East
3974	ORD103999	2023-01-20	CUST1421	Ayu Lee	South
3975	ORD104000	2024-02-12	CUST1580	Linda Tan	Midwest
3976					
3977					
3978					
3979					
3980				customer_name	
3981				D3975)	
3982					
3983					
3984					
3985					

Figure 21. The COUNTBLANK formula applied outside the table to count missing values per column

customer_name		120
region		79
payment_method		158
order_date		0
quantity		0
unit_price		77

Figure 22. Initial tally — order_date briefly read as 0 because a blank value was hidden as the number 0

customer_name		120
region		79
payment_method		158
order_date		0
quantity		59
unit_price		77

Figure 23. Tally after the quantity values that had been disguised as 0 were reverted to blank (rising to 59)

order_date briefly read as 0 missing, even though a value was hidden as the number 0 (a former blank cell formatted as the date January 1, 1900). It was found by filtering the order_date column by year:

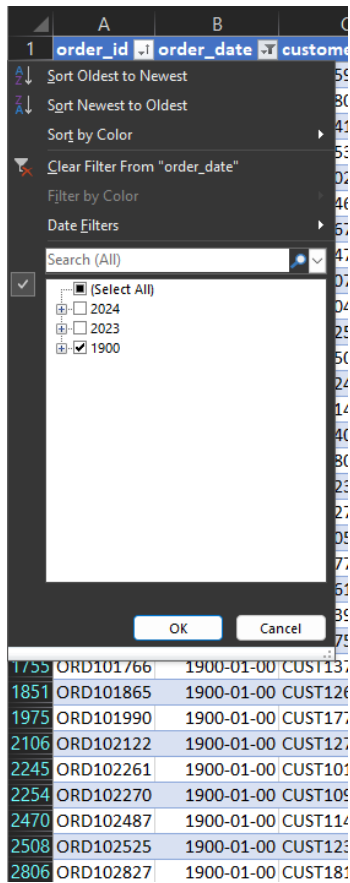


Figure 24. Filtering order_date uncovered rows with the year 1900 — a hidden former blank cell

customer_name		120
region		79
payment_method		158
order_date		40
quantity		59
unit_price		77

Figure 25. Final missing-value tally per column after every hidden value was found

The handling strategy was tailored per column — categorical columns were filled with an "Unknown" label, columns used in calculations (quantity, unit_price) were filled using statistical estimates, while order_date rows were removed because they could not be reliably reconstructed.

Column	Count	%	Handling Strategy
payment_method	158	4.0%	Filled with "Unknown" — a categorical column, does not affect any calculation.
customer_name	120	3.0%	Filled with "Unknown Customer" — identity is still available via customer_id.
region	79	2.0%	Recovered from other transactions by the same customer_id (INDEX/MATCH array formula); the remainder filled with "Unknown".
unit_price	77	1.9%	Filled with the average price per product_name (AVERAGEIFS) — more accurate than a global average.
quantity	59	1.5%	Filled with the overall median quantity (1 unit) — median chosen because the data skews toward small numbers.
order_date	40	1.0%	Rows removed — transaction dates are critical for trend analysis and cannot be reliably reconstructed.

Table 2. Summary of missing values and their handling strategy

payment_method and customer_name

Both categorical columns were filled by filtering for (Blanks), then Select All + typing the replacement value + Alt+Enter to fill every selected cell at once:

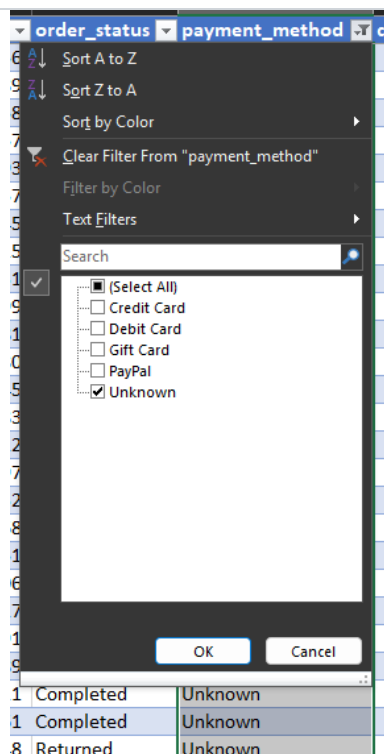


Figure 26. Filtering payment_method to show blank rows, ready to be filled with "Unknown"

region — recovery attempted first

The region column was first recovered from other transactions by the same customer_id before falling back to "Unknown", using an array formula (entered with Ctrl+Shift+Enter):

```
=IFERROR(INDEX($region$2:$region$3975, MATCH(1,
($customer_id$2:$customer_id$3975=[@customer_id]) * ($region$2:$region$3975<>""), 0)),
"") {Ctrl+Shift+Enter}
```

	A	B	C	D	E	F	G	H	I	J	K	L
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	region_recovered
1	ORD100553	2024-11-04	CUST1305	Linda Nugroho		Home & Kitchen	Storage Container	1	31.12	Completed	Debit Card	East
2	ORD101005	2023-04-05	CUST1755	Budi Sari		Electronics	Smartphone Case	1	29.75	Returned	PayPal	East
3	ORD101116	2023-11-02	CUST1733	Lisa Sari		Beauty	Face Mask Set	1	29.15	Cancelled	Debit Card	East
4	ORD101125	2023-01-27	CUST1516	Fajar Brown		Electronics	Webcam	1		Returned	Gift Card	East
5	ORD101190	2023-03-17	CUST1638	Maria Kurniawan		Toys	Building Blocks	1	29.17	Completed	Credit Card	East
6	ORD101319	2023-01-24	CUST1707	Michael Yusuf		Electronics	Power Bank	3	70.28	Completed	Debit Card	East
7	ORD101435	2023-05-15	CUST1844	Ahmad Johnson		Clothing	Baseball Cap	5	59.21	Completed	PayPal	East
8	ORD101524	2024-01-18	CUST1172	Nina Lee		Home & Kitchen	Blender	1	78.32	Completed	Gift Card	East
9	ORD101662	2024-04-08	CUST1121	David Nugroho		Home & Kitchen	Storage Container	1	24.34	Completed	PayPal	East
10	ORD102055	2024-08-04	CUST1708	Ayu Smith		Beauty	Face Mask Set	2	32.63	Completed	Gift Card	East
11	ORD102156	2024-01-11	CUST1798	Siti Anderson		Beauty	Lip Balm	1	6.38	Completed	Gift Card	East
12	ORD102170	2023-10-16	CUST1561	Nina Santos		Toys	Building Blocks	2	41.55	Completed	Debit Card	East
13	ORD102589	2024-05-25	CUST1558	Sarah Wilson		Sports & Outdoors	Hiking Backpack	2	56.86	Completed	Credit Card	East
14	ORD102759	2024-12-22	CUST1572	Maria Yusuf		Toys	Board Game	2	52.88	Cancelled	Credit Card	East
15	ORD102912	2024-03-14	CUST1851	Robert Salim		Toys	Action Figure	4	33.78	Completed	Gift Card	East
16	ORD103206	2024-10-17	CUST1417	Unknown Customer		Electronics	Webcam	1		Completed	PayPal	East
17	ORD103366	2024-11-14	CUST1723	Ayu Wijaya		Home & Kitchen	LED Desk Lamp	2	43.94	completed	Debit Card	East
18	ORD103436	2023-07-04	CUST1402	Rina Nugroho		Clothing	Wool Socks	1	33.39	Completed	PayPal	East

Figure 27. The region_recovered column successfully found a region from another transaction by the same customer

unit_price and quantity — filled with statistical estimates

The unit_price column was filled using AVERAGEIFS — the average price computed specifically from the same product (product_name), not the average across the whole dataset:

```
=IF([@unit_price]="", AVERAGEIFS($unit_price$2:$unit_price$3975,
$product_name$2:$product_name$3975, [@product_name]), [@unit_price])
```

unit_price	order_status	payment_method	unit_price_fill
	Completed	Credit Card	83.09465909
	Pending	Gift Card	63.9183908
	Completed	Credit Card	36.80701923
	Completed	PayPal	63.9183908

Figure 28. The unit_price_fill column — blank prices filled with the average price of the same product

The quantity column was filled using MEDIAN (not AVERAGE), because the quantity data skews toward small numbers (1–5), making the median more resistant to distortion than the average:

```
=IF([@quantity]="", MEDIAN($quantity$2:$quantity$3975), [@quantity])
```

H	I	J	K	L	M
quantity	unit_price	order_status	payment_method	unit_price_fill	quantity_fill
	21.69	Completed	Credit Card	21.69	1
	11.03	Completed	Gift Card	11.03	1
	58.91	Completed	Credit Card	58.91	1
	122.70	Completed	Debit Card	122.7	1
	17.39	Completed	Credit Card	17.39	1
	49.25	Returned	Credit Card	49.25	1
	13.42	Completed	Debit Card	13.42	1
	64.16	Completed	Unknown	64.16	1
	33.37	Completed	Gift Card	33.37	1
	58.85	Completed	Gift Card	58.85	1
	19.42	Completed	Debit Card	19.42	1
	76.05	Completed	Debit Card	76.05	1
	75.06	Completed	Debit Card	75.06	1
		Completed	Credit Card	83.09465909	1
	17.66	Completed	Debit Card	17.66	1

Figure 29. The unit_price_fill and quantity_fill columns shown together — blank quantities consistently filled with the median (1)

Quick reference — the basic syntax of these three functions:

- AVERAGE(range) — the average of every number in the range, with no condition.
- AVERAGEIFS(average_range, criteria_range1, criteria1, ...) — the average of average_range, but only for rows meeting the criteria (more than one criterion is allowed).
- MEDIAN(range) — the middle value of a set of numbers; more resistant to outliers than AVERAGE.

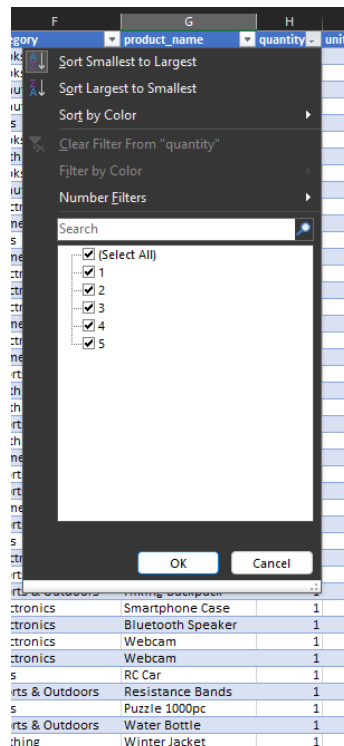


Figure 30. The quantity filter after every missing value was handled — only reasonable values of 1–5 remain

Merging the results back into the original column

When moving a helper column's results (such as region_recovered) back into the original column, a more efficient way to handle a large number of rows is to sort (Sort A-Z) the helper column first — successfully

recovered rows will cluster into one group, allowing them to be copied group by group instead of one at a time:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	unit_price_fill	quantity_fill
29	ORD103119		CUST1171	James Anderson	East	Beauty	Shampoo	1	8.19	Completed	PayPal	8.19	1
333	ORD101366		CUST1407	Rina Smith	East	Books	Children's Book	2	13.80	Returned	PayPal	13.80	2
586	ORD100368		CUST1472	Andi Halim	Midwest	Beauty	Shampoo	2	17.57	Completed	Gift Card	17.57	2
625	ORD103834		CUST1853	Robert Smith	East	Beauty	Face Mask Set	2	18.12	Completed	Credit Card	18.12	2
830	ORD102827		CUST1818	Siti Salim	East	Toys	RC Car	1	21.00	Completed	Credit Card	21.00	1
845	ORD102261		CUST1016	John Pratama	Midwest	Beauty	Lip Balm	1	21.12	Completed	Debit Card	21.12	1
947	ORD101212		CUST1503	Andi Anderson	East	Beauty	Sunscreen SPF50	1	22.65	Returned	Gift Card	22.65	1
987	ORD100583		CUST1076	Kevin Santos	East	Clothing	Running Shorts	2	23.24	Completed	PayPal	23.24	2
1058	ORD100121		CUST1538	Dewi Kurniawan	East	Toys	RC Car	1	24.15	Returned	Gift Card	24.15	1
1074	ORD100824		CUST1040	Lisa Garcia	East	Toys	Puzzle 1000pc	2	24.38	Cancelled	PayPal	24.38	2
1098	ORD101865		CUST1267	Maria Anderson	East	Home & Kitchen	Blender	2	24.78	Completed	PayPal	24.78	2
1278	ORD100265		CUST1023	Siti Wilson	East	Home & Kitchen	Non-Stick Pan	1	28.43	Completed	Debit Card	28.43	1
1500	ORD101196		CUST1254	Nina Garcia	East	Beauty	Sunscreen SPF50	2	30.21	returned	Gift Card	30.21	2
1649	ORD101990		CUST1173	Andi Nugroho	Midwest	Sports & Outdoors	Resistance Bands	2	32.18	Completed	Debit Card	32.18	2
1670	ORD101307		CUST1248	Andi Pratama	East	Toys	Board Game	1	32.42	Completed	Credit Card	32.42	1
1833	ORD100317		CUST1674	Dewi Lee	East	Sports & Outdoors	Resistance Bands	1	35.10	Completed	PayPal	35.10	1
1940	ORD102270		CUST1093	Kevin Nugroho	Midwest	Electronics	Smartphone Case	1	37.47	Completed	PayPal	37.47	1
2167	ORD101408		CUST1808	Michael Kurniawan	Midwest	Home & Kitchen	Coffee Maker	1	41.66	Pending	Credit Card	41.66	1
2283	ORD103651		CUST1376	Nina Yusuf	Midwest	Toys	Action Figure	2	44.14	Completed	Gift Card	44.14	2
2300	ORD101765		CUST1157	Sarah Lee	East	Beauty	Moisturizer	1	44.45	Cancelled	Debit Card	44.45	1
2677	ORD103515		CUST1178	Kevin Wilson	East	Home & Kitchen	Throw Pillow	2	54.89	completed	Gift Card	54.89	2
2707	ORD100294		CUST1462	Michael Lee	Midwest	Toys	RC Car	4	55.58	Completed	Credit Card	55.58	4
2917	ORD101628		CUST1050	Sarah Anderson	East	Sports & Outdoors	Jump Rope	2	61.56	Completed	Debit Card	61.56	2
2934	ORD103310		CUST1374	Nina Garcia	Midwest	Clothing	Cotton T-Shirt	1	62.48	Completed	Credit Card	62.48	1
3108	ORD101766		CUST1379	Putri Halim	East	Sports & Outdoors	Resistance Bands	2	69.99	Completed	Debit Card	69.99	2
3113	ORD101704		CUST1398	Nina Kurniawan	East	Electronics	Smartphone Case	1	70.07	Completed	PayPal	70.07	1
3120	ORD101325		CUST1149	Sarah Yusuf	Midwest	Clothing	Cotton T-Shirt	1	70.18	Completed	Gift Card	70.18	1
3167	ORD102122		CUST1278	David Salim	East	Sports & Outdoors	Camping Tent	2	72.10	Completed	PayPal	72.10	2
3260	ORD101426		CUST1128	David Salim	East	Electronics	USB-C Cable	1	76.10	Pending	PayPal	76.10	1
3301	ORD101644		CUST1774	Maria Wijaya	East	Home & Kitchen	Non-Stick Pan	1	77.96	completed	PayPal	77.96	1
3377	ORD100056		CUST1415	Sarah Salim	East	Sports & Outdoors	Hiking Backpack	1	82.78	Completed	Credit Card	82.78	1
3516	ORD101650		CUST1616	Putri Wijaya	Midwest	Electronics	Power Bank	1	91.79	Completed	PayPal	91.79	1
3551	ORD100014		CUST1590	Putri Tan	East	Home & Kitchen	LED Desk Lamp	3	93.60	Completed	Gift Card	93.60	3
3573	ORD100037		CUST1806	Ahmad Hidayat	East	Electronics	Wireless Mouse	1	94.76	Completed	PayPal	94.76	1
3618	ORD101425		CUST1238	John Pratama	East	Electronics	Wireless Earbuds	2	97.63	Completed	Debit Card	97.63	2
3706	ORD102525		CUST1233	Michael Lee	East	Electronics	Smartphone Case	2	104.81	Returned	Debit Card	104.81	2
3734	ORD102969		CUST1816	John Kurniawan	East	Home & Kitchen	Coffee Maker	1	107.95	Completed	Gift Card	107.95	1
3738	ORD103137		CUST1086	Rina Anderson	West	Home & Kitchen	Non-Stick Pan	2	108.13	Pending	PayPal	108.13	2
3778	ORD102975		CUST1246	Ayu Brown	East	Electronics	Laptop Stand	1	112.19	Completed	Debit Card	112.19	1
3896	ORD102487		CUST1143	Linda Wijaya	East	Electronics	Laptop Stand	1	129.86	Completed	PayPal	129.86	1
3976													

Figure 31. The region_recovered column sorted A-Z to make copying results group by group into the original column easier

- order_date: 40 blank rows were removed because transaction dates are critical for trend analysis and cannot be reliably reconstructed.

3.6 Creating a Calculated Column (Revenue)

The revenue column was added as a simple multiplication of quantity and unit_price, so its value is automatically consistent with all the preceding cleaning steps:

`=[@quantity]*[@unit_price]`

H	I	J	K	L	M
quantity	unit_price	order_status	payment_method	quantity_fill	revenue
1	6.00	Completed	Gift Card	1	6
1	6.22	Completed	Gift Card	1	6.22
2	6.22	Completed	Credit Card	2	12.44
2	6.27	Completed	Debit Card	2	12.54
1	6.38	Completed	Gift Card	1	6.38
4	6.40	Completed	Debit Card	4	25.6
2	6.72	Completed	Gift Card	2	13.44
1	6.80	Returned	PayPal	1	6.8
2	6.84	Returned	PayPal	2	13.68
2	6.86	Completed	Credit Card	2	13.72
4	6.88	Completed	Debit Card	4	27.52

Figure 32. The revenue column on the first rows right after the formula was applied

H	I	J	K	L	M	N	O
quantity	unit_price	order_status	payment_method	quantity_fill	revenue		
1	6.00	Completed	Gift Card	1	6.00		
1	6.22	Completed	Gift Card	1	6.22		
2	6.22	Completed	Credit Card	2	12.44		
2	6.27	Completed	Debit Card	2	12.54		
1	6.38	Completed	Gift Card	1	6.38		
4	6.40	Completed	Debit Card	4	25.60		
2	6.72	Completed	Gift Card	2	13.44		
1	6.80	Returned	PayPal	1	6.80		
2	6.84	Returned	PayPal	2	13.68		
2	6.86	Completed	Credit Card	2	13.72		
1	6.92	Completed	PayPal	1	6.92		
2	6.95	Completed	Debit Card	2	13.90		
1	6.96	Completed	Debit Card	1	6.96		
1	6.98	Completed	Gift Card	1	6.98		
1	6.98	Completed	Gift Card	1	6.98		
1	7.06	Pending	Credit Card	1	7.06		
2	7.18	Completed	PayPal	2	14.36		
4	7.19	Completed	Gift Card	4	28.76		
1	7.25	Completed	Gift Card	1	7.25		
1	7.26	Completed	Gift Card	1	7.26		
4	7.44	Completed	PayPal	4	29.76		
2	7.83	Completed	PayPal	2	15.66		
1	7.84	Completed	Gift Card	1	7.84		
1	7.88	Completed	Debit Card	1	7.88		
5	7.95	Completed	Credit Card	5	39.75		
1	8.13	Pending	Debit Card	1	8.13		
2	8.17	Completed	Credit Card	2	16.34		
3	8.20	Completed	Debit Card	3	24.60		
2	8.22	Completed	Debit Card	2	16.44		
3	8.23	Cancelled	Gift Card	3	24.69		
1	8.29	Completed	Gift Card	1	8.29		
1	8.29	Completed	Debit Card	1	8.29		
1	8.33	Completed	Gift Card	1	8.33		
1	8.39	Completed	Credit Card	1	8.39		
1	8.41	Completed	Gift Card	1	8.41		
1	8.41	Completed	Credit Card	1	8.41		
1	8.41	Pending	Credit Card	1	8.41		
1	8.43	Completed	Debit Card	1	8.43		
3	8.47	Returned	Credit Card	3	25.41		
1	8.49	Completed	Gift Card	1	8.49		
1	8.56	Pending	Gift Card	1	8.56		
1	8.59	Completed	PayPal	1	8.59		
3	8.64	Completed	PayPal	3	25.92		
1	8.75	completed	Debit Card	1	8.75		
1	8.75	Completed	Gift Card	1	8.75		
1	8.87	Completed	Credit Card	1	8.87		
1	8.88	completed	Gift Card	1	8.88		
1	8.90	Completed	PayPal	1	8.90		
1	8.91	Pending	Credit Card	1	8.91		
2	8.97	Completed	Gift Card	2	17.94		
2	8.98	Completed	Gift Card	2	17.96		
1	9.04	Completed	Gift Card	1	9.04		
1	9.04	Completed	Credit Card	1	9.04		
2	9.04	Completed	Debit Card	2	18.08		
1	9.05	Completed	Credit Card	1	9.05		
4	9.05	Completed	PayPal	4	36.20		
1	9.05	Completed	Unknown	1	9.05		
1	9.06	Completed	Debit Card	1	9.06		
1	9.06	Completed	PayPal	1	9.06		
3	9.07	Completed	Debit Card	3	27.21		
1	9.08	Completed	PayPal	1	9.08		

Figure 33. The revenue column on the dataset — total revenue \$306,888.66 across 3,934 rows (before 1 additional duplicate was found during the re-verification in section 3.1)

Hari	Fase	Task	Tools	Selesai?
Day 1	Excel Data Cleaning	Ambil dataset e-commerce sales	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek & tangani missing values	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek & hapus duplicate records	Excel	SELESAI
Day 1	Excel Data Cleaning	Perbaiki data types yang salah	Excel	SELESAI
Day 1	Excel Data Cleaning	Standarkan kategori yang tidak konsisten	Excel	SELESAI
Day 1	Excel Data Cleaning	Format ulang kolom tanggal	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek outlier	Excel	SELESAI
Day 1	Excel Data Cleaning	Buat calculated columns	Excel	SELESAI
Day 1	Excel Data Cleaning	Dokumentasikan raw -> cleaning -> clean dataset	Excel	
Day 1	Excel Data Cleaning	Simpan sebagai cleaned_sales_data.xlsx	Excel	

Figure 34. Day 1 checklist — every cleaning step complete, only documentation and saving the file remain

4. Results Summary (Cleaning Log)

The table below summarizes every cleaning step in order, and is included as a separate sheet ("Cleaning Log") in the cleaned_sales_data.xlsx file as an audit trail of the process.

Step	Rows Affected	Notes
1. Remove duplicate records	99 rows	Exact duplicates (all columns identical), presumably double data entry — 98 rows in the initial cleanup + 1 row found during re-verification on Day 2 (row-signature + color-filter technique).
2. Standardize category & region	—	Standardized to Proper Case; typos fixed (Cloths→Clothing, Electronic→Electronics, Mid-West→Midwest).
3. Standardize order_date	—	5 mixed date formats consolidated into YYYY-MM-DD.

4. Remove quantity outliers	15 rows	Extreme values (250/500/999) implausible for individual retail transactions, scattered randomly with no wholesale pattern.
5. Remove unit_price outliers	12 rows	Extreme values (8888/4999.99/1234.56) inconsistent with product category; identical repeated values suggest a system error.
6. Handle missing values	533 cells	See Table 2 — strategy differs per column based on importance and recoverability.
7. Create the revenue calculated column	3,933 rows	revenue = quantity × unit_price. Dataset total revenue: \$306,888.66.

Table 3. Cleaning Log — summary of the entire raw dataset → clean dataset process

5. Conclusion

The final dataset contains 3,933 clean rows, with no missing values, and consistent data types and formatting across every column. A total of 167 rows (4.1% of the original data) were removed throughout the process, all through documented and justifiable decisions rather than arbitrary deletion. One additional duplicate that slipped through the initial cleanup was found through re-verification before the data was used for analysis — an important quality-assurance practice to carry out before a dataset is considered final.

The output of this stage is the `cleaned_sales_data.xlsx` file, containing two sheets: the cleaned transaction data, and a Cleaning Log documenting the process. This dataset becomes the single source used consistently throughout the following stages: Excel analysis (Day 2), SQL queries (Day 3), exploratory data analysis with Python (Day 4), and the Tableau dashboard (Day 5).